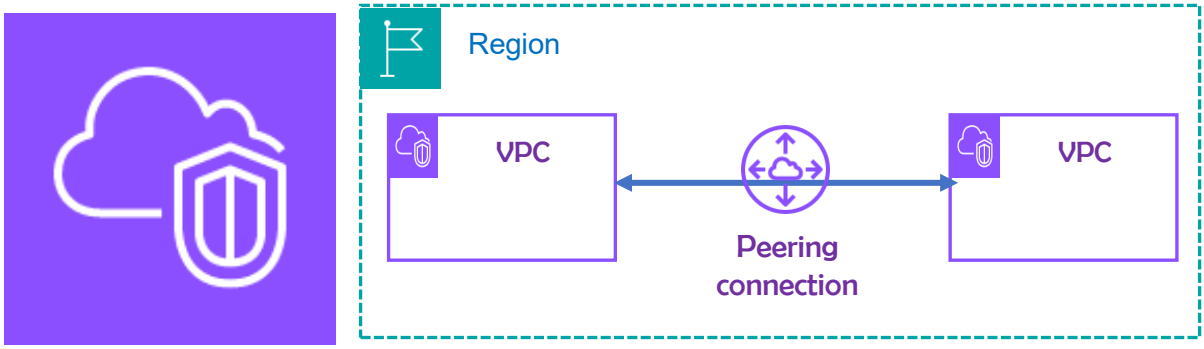
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Capstone Projects

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Configure VPC Peering within the Region in AWS



GOKUL S

Objective

Amazon VPC Peering is a networking connection between two Virtual Private Clouds (VPCs) that enables you to route traffic between them privately using AWS's internal network.

🔷 Key Features:

✅ Private communication (no internet gateway required)

✅ Low latency, high bandwidth

✅ Works across regions (Inter-Region Peering)

✅ No single point of failureSubnets:

VPC peering is a one-to-one networking connection between two VPCs that enables you to route traffic between them privately. Instances in either VPC can communicate with each other like they are in the same network. You can create a VPC peering connection between your own VPCs, with a VPC in another AWS account, or with a VPC in a different AWS Region.

You can establish peering relationships between VPCs across different AWS Regions. Inter-Region VPC peering provides a simple and cost-effective way to share resources between Regions or replicate data for geographic redundancy. Data that is transferred across inter-Region VPC peering connections is charged at the standard inter-Region data transfer rates.

Inter-Region VPC peering enables VPC resources to communicate with each other using private IP addresses without requiring gateways, VPN connections, or separate network appliances. Some examples of VPC resources include Amazon Elastic Compute Cloud (Amazon EC2) instances, Amazon Relational Database Service (Amazon RDS) databases, and AWS Lambda functions that run in different Regions.

Traffic remains in the private IP address space. All inter-Region traffic is encrypted with no single point of failure or bandwidth bottleneck. Traffic always stays on the global AWS backbone. Traffic never traverses the public internet, which reduces threats, such as common exploits and distributed denial of service (DDoS) attacks.

🔷 Use Cases:

✔️ Connecting microservices across VPCs

✔️ Multi-account architecture

✔️ Hybrid application environments

✔️ Data sharing between teams

🔷 Important Limitations:

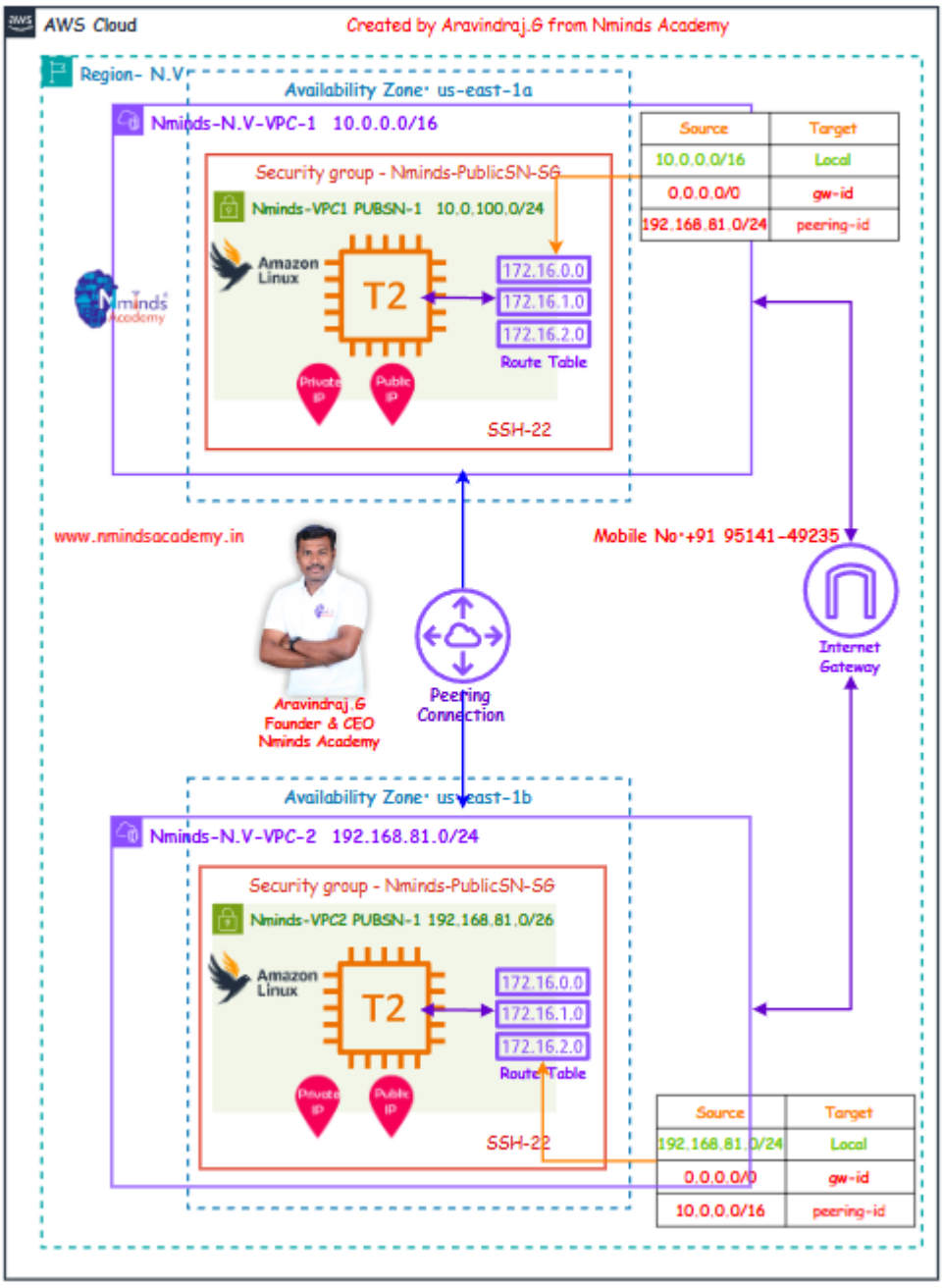
⚠️ No transitive peering (VPC A ↔ B ↔ C does NOT mean A ↔ C)

⚠️ Overlapping CIDR blocks are not allowed

⚠️ Must manually update route tables

Topology

How to Configure VPC Peering within the Region in AWS



Used Services in AWS



Amazon VPC

Public subnet

Private subnet



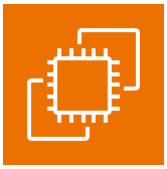
Internet gateway



Route table



Peering connection



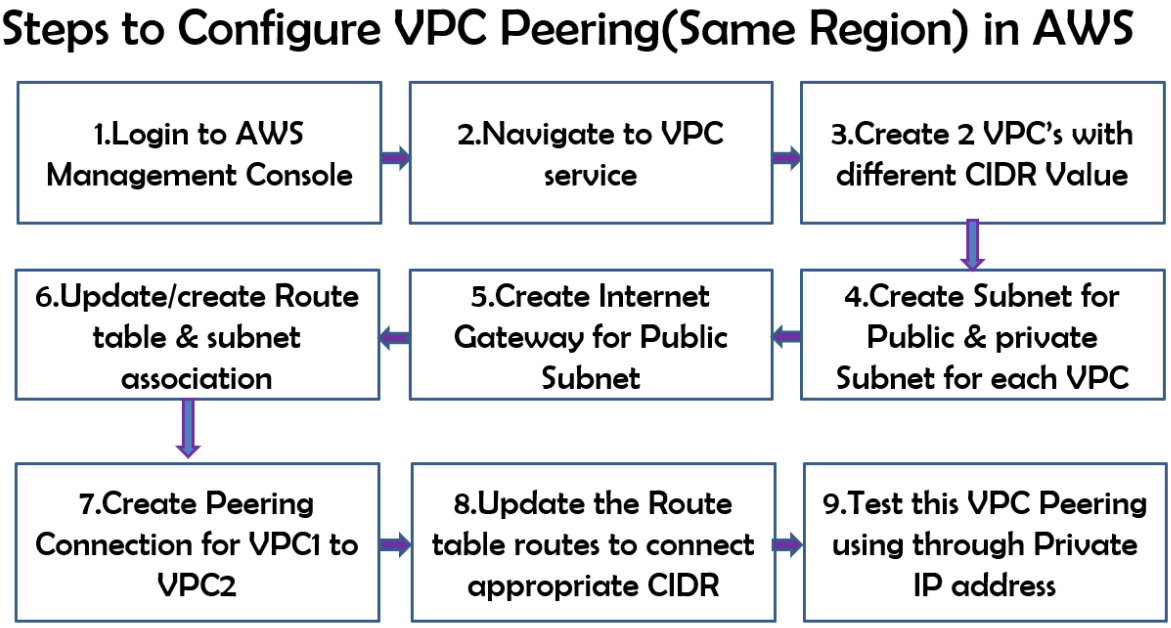
Amazon Elastic Compute Cloud (Amazon EC2)

Used Services in Windows Server



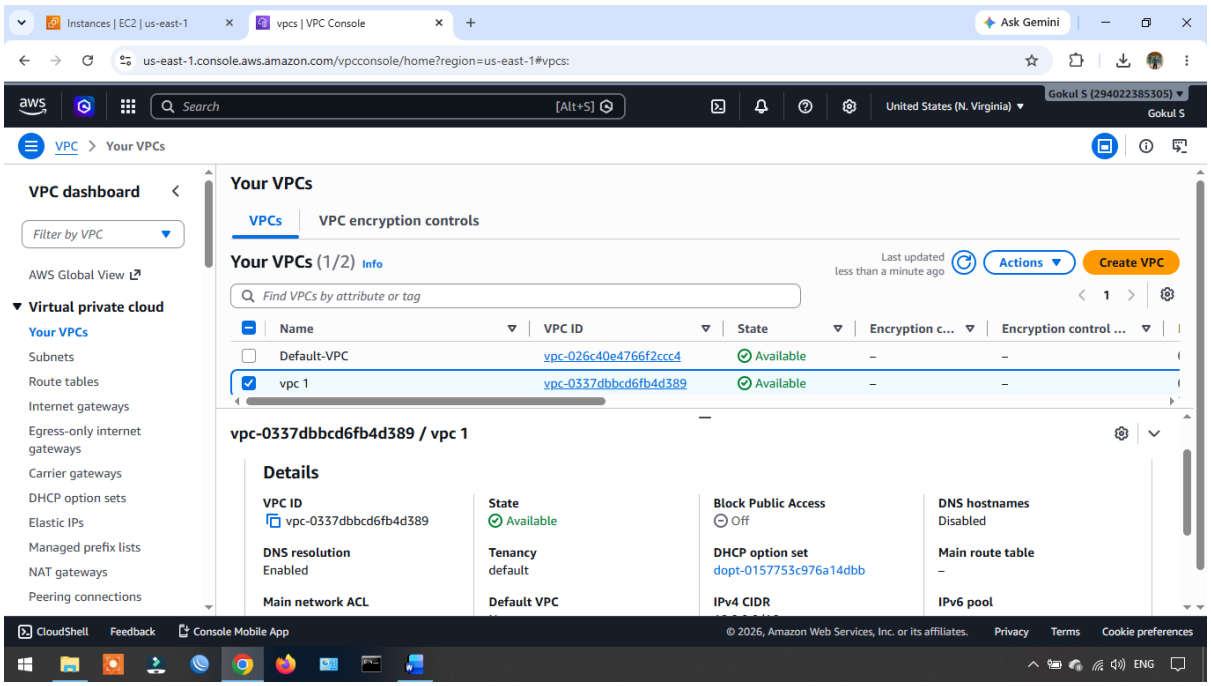
Apache Webserver

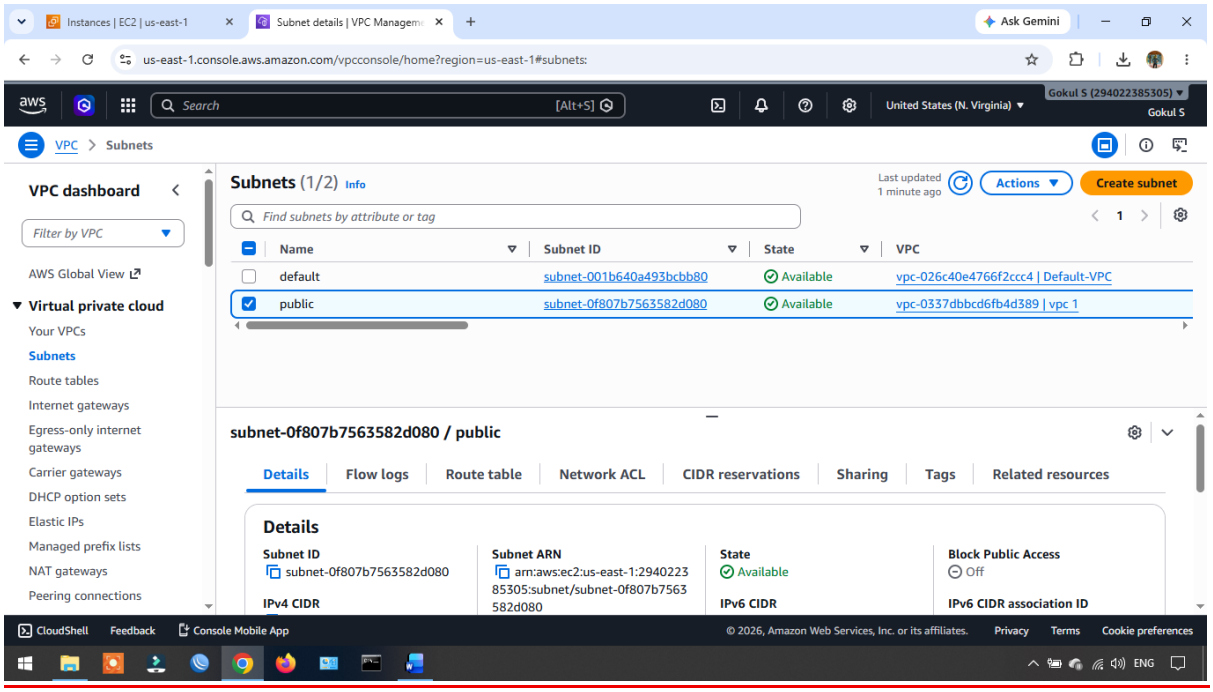
Execution Steps:

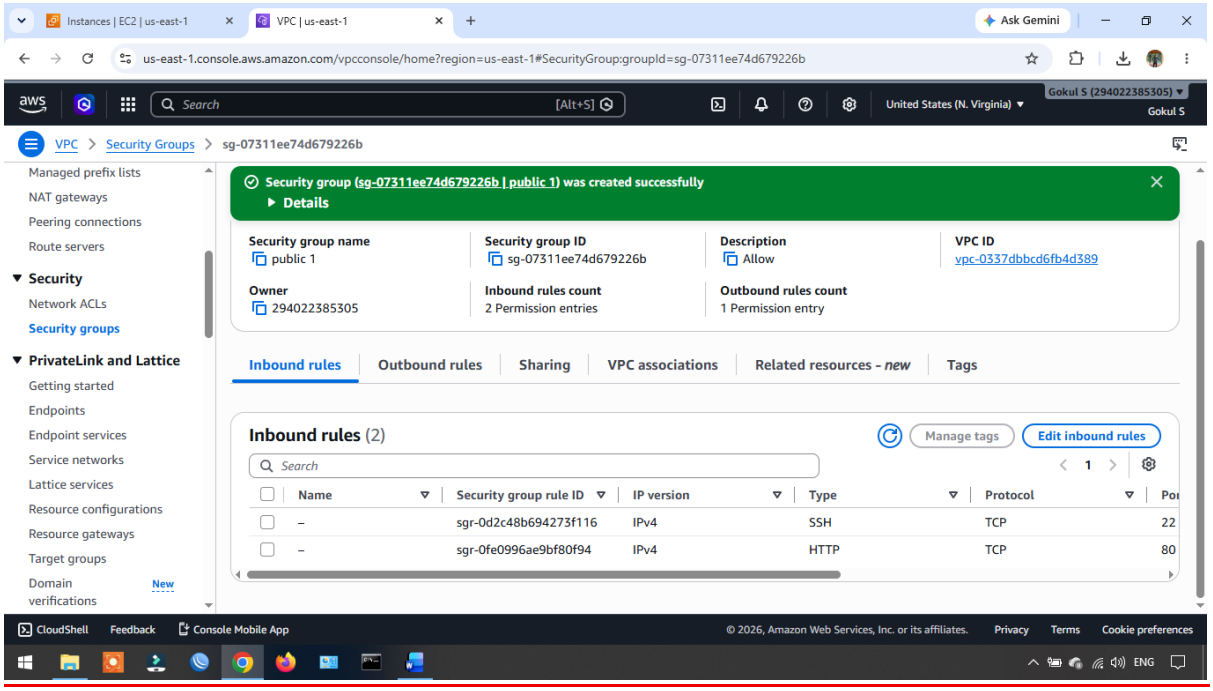


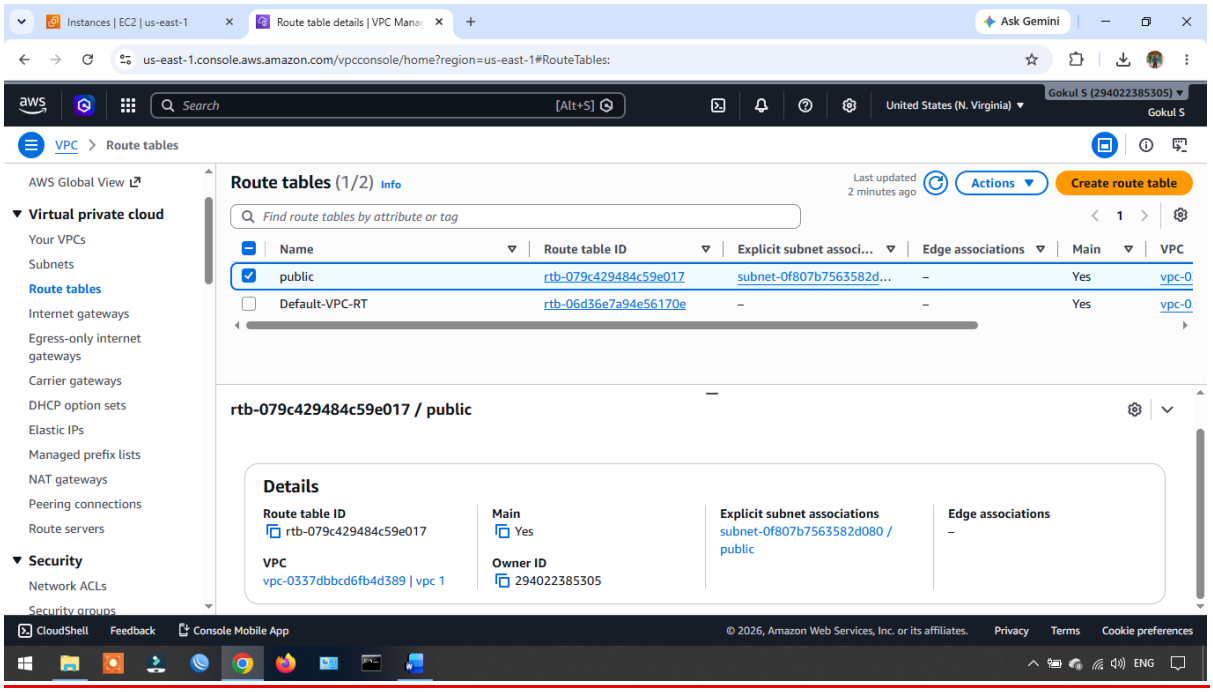
Execution Tasks:

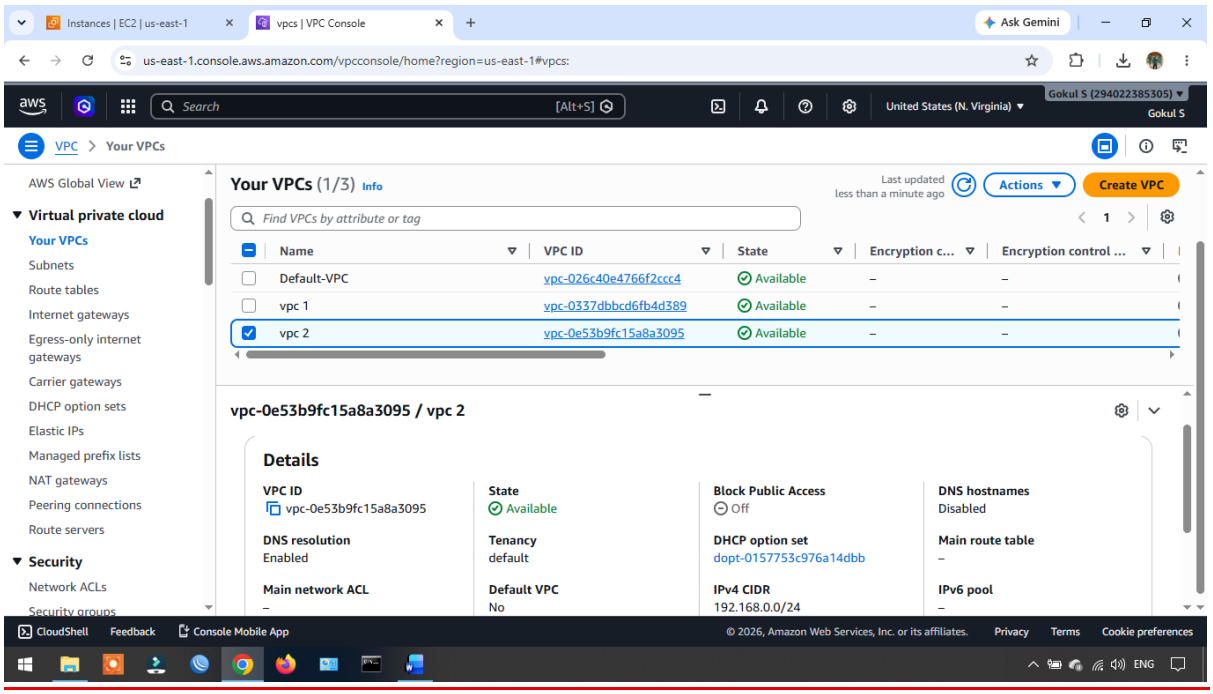
Task1: Create the 2 custom VPC with Public Subnet with different VPC based on the above topology:

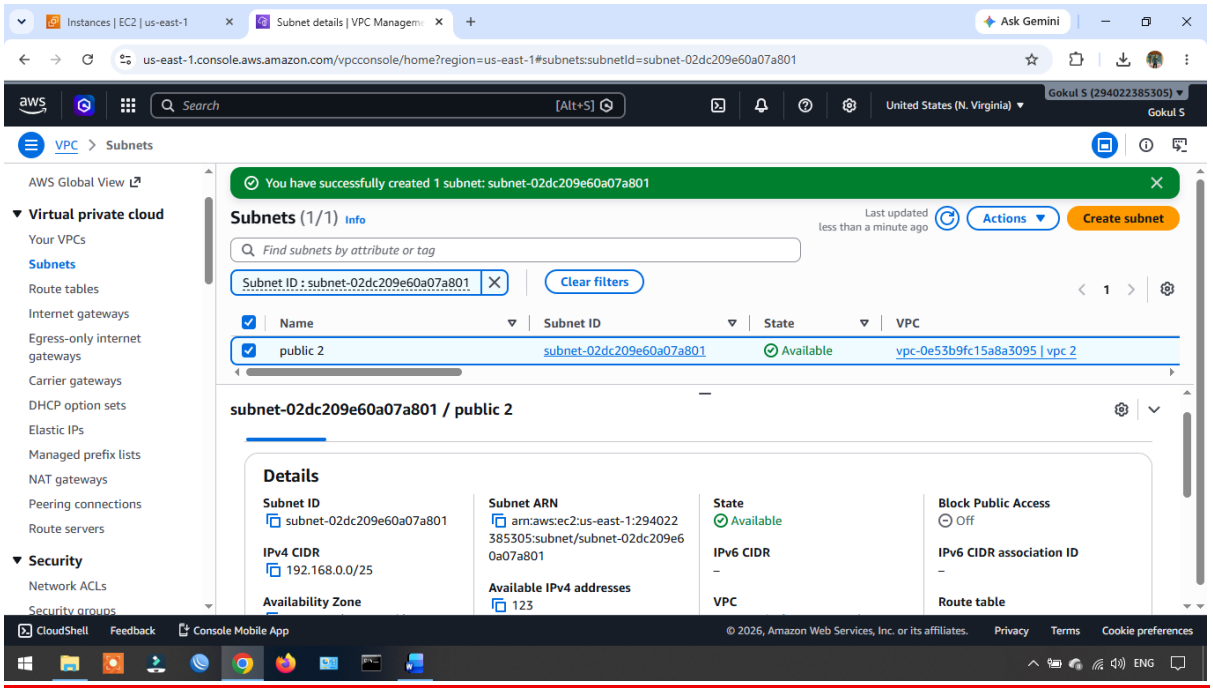


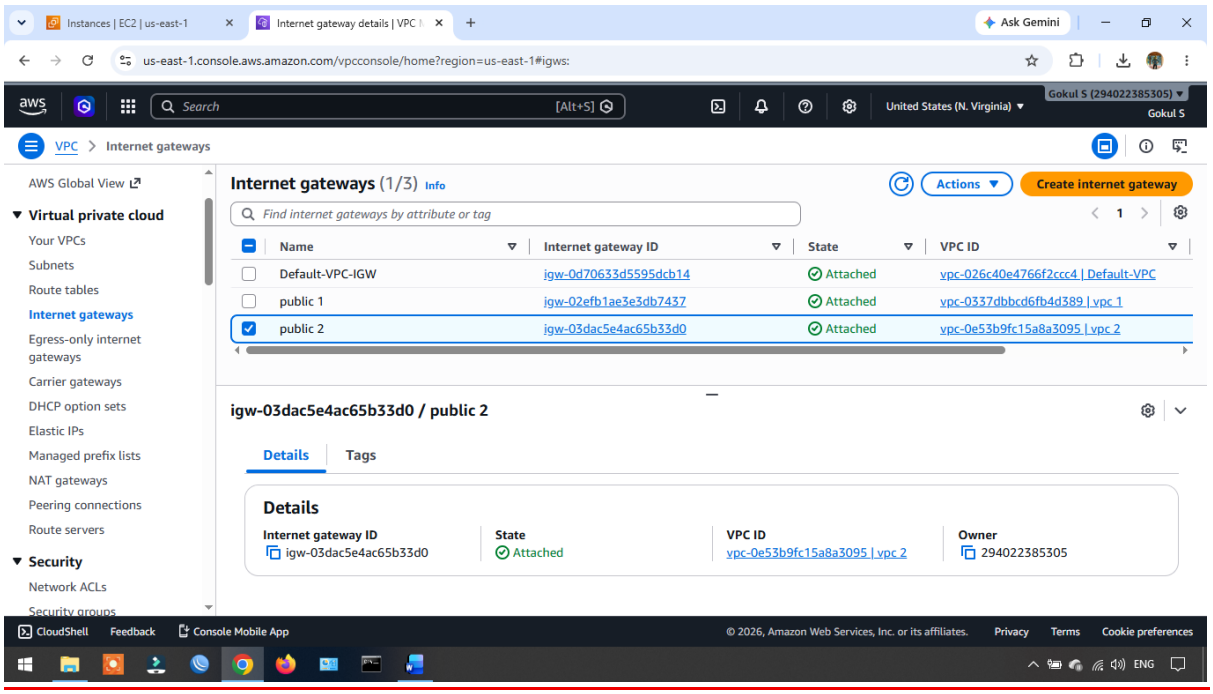

Instances | EC2 | us-east-1
Subnet details | VPC Management
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#subnets:
Search
[Alt+S]
United States (N. Virginia)
Gokul S (294022385305)
Gokul S
VPC > Subnets
VPC dashboard
Filter by VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Subnets (1/2) Info
Last updated 1 minute ago
Actions
Create subnet
Find subnets by attribute or tag
Name
Subnet ID
State
VPC
default
subnet-001b640a493bcbb80
Available
vpc-026c40e4766f2ccc4 | Default-VPC
public
subnet-0f807b7563582d080
Available
vpc-0337dbbcd6fb4d389 | vpc 1
subnet-0f807b7563582d080 / public
Details
Flow logs
Route table
Network ACL
CIDR reservations
Sharing
Tags
Related resources
Details
Subnet ID
subnet-0f807b7563582d080
Subnet ARN
arn:aws:ec2:us-east-1:294022385305:subnet/subnet-0f807b7563582d080
State
Available
Block Public Access
Off
IPv4 CIDR
IPv6 CIDR
IPv6 CIDR association ID
CloudShell
Feedback
Console Mobile App
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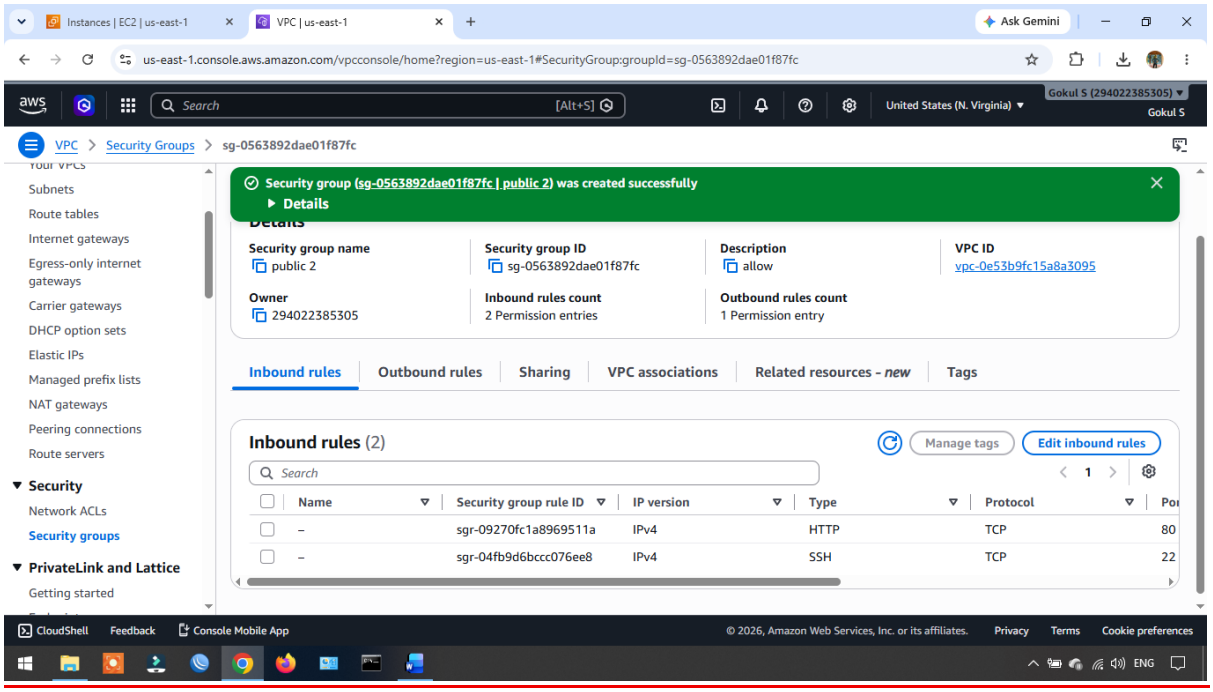

Instances | EC2 | us-east-1
VPC | us-east-1
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#SecurityGroup:groupId=sg-07311ee74d679226b
Search
[Alt+S]
United States (N. Virginia)
Gokul S (294022385305)
Gokul S
VPC > Security Groups > sg-07311ee74d679226b
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Lattice services
Resource configurations
Resource gateways
Target groups
Domain verifications
New
Security group (sg-07311ee74d679226b | public 1) was created successfully
Details
Security group name
public 1
Security group ID
sg-07311ee74d679226b
Description
Allow
VPC ID
vpc-0337dbbcd6fb4d389
Owner
294022385305
Inbound rules count
2 Permission entries
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (2)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
sgr-0d2c48b694273f116
IPv4
SSH
TCP
22
sgr-0fe0996ae9bf80f94
IPv4
HTTP
TCP
80
CloudShell
Feedback
Console Mobile App
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Instances | EC2 | us-east-1
Route table details | VPC Mana
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTables:
Search
[Alt+S]
United States (N. Virginia)
Gokul S (294022385305)
Gokul S
VPC > Route tables
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
Route tables (1/2) Info
Last updated 2 minutes ago
Actions
Create route table
Find route tables by attribute or tag
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
public
rtb-079c429484c59e017
subnet-0f807b7563582d...
–
Yes
vpc-0
Default-VPC-RT
rtb-06d36e7a94e56170e
–
–
Yes
vpc-0
rtb-079c429484c59e017 / public
Details
Route table ID
rtb-079c429484c59e017
Main
Yes
Explicit subnet associations
subnet-0f807b7563582d080 / public
Edge associations
–
VPC
vpc-0337dbbcd6fb4d389 | vpc 1
Owner ID
294022385305
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Feedback
Console Mobile App
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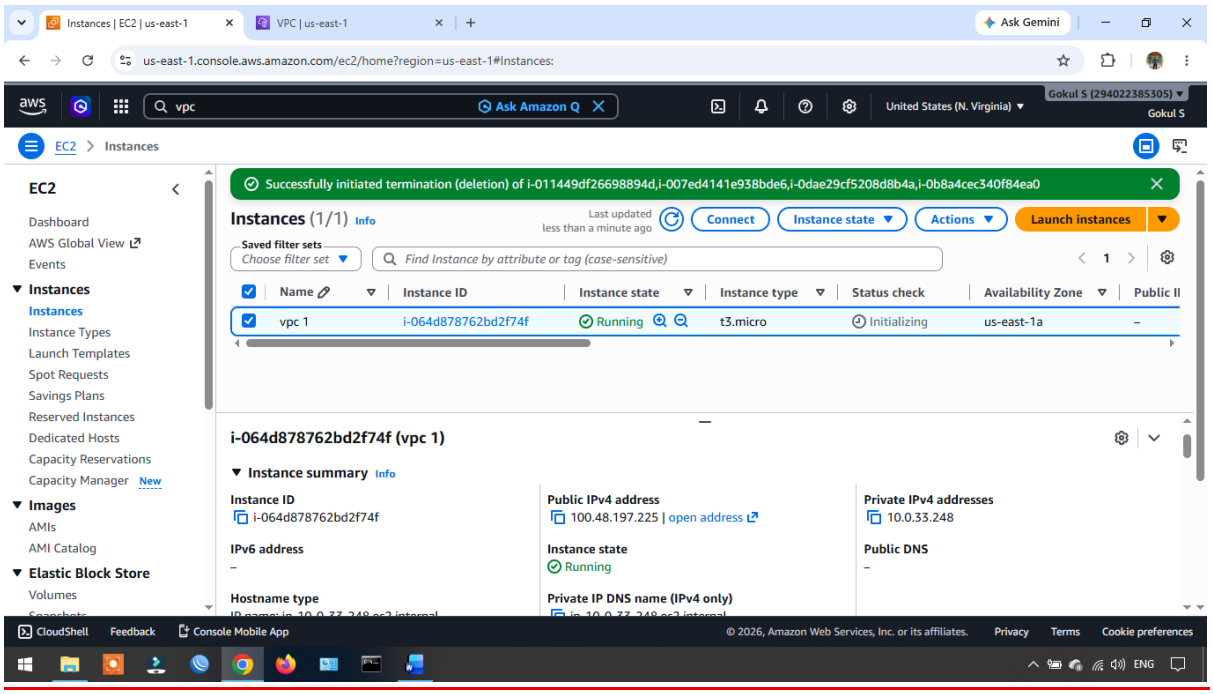

Instances | EC2 | us-east-1
vpcs | VPC Console
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#vpcs:
Search
[Alt+S]
United States (N. Virginia)
Gokul S (294022385305)
Gokul S
VPC > Your VPCs
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
Your VPCs (1/3) Info
Last updated less than a minute ago
Actions
Create VPC
Find VPCs by attribute or tag
Name
VPC ID
State
Encryption c...
Encryption control ...
Default-VPC
vpc-026c40e4766f2ccc4
Available
vpc 1
vpc-0337dbbcd6fb4d389
Available
vpc 2
vpc-0e53b9fc15a8a3095
Available
vpc-0e53b9fc15a8a3095 / vpc 2
Details
VPC ID
vpc-0e53b9fc15a8a3095
State
Available
Block Public Access
Off
DNS hostnames
Disabled
DNS resolution
Enabled
Tenancy
default
DHCP option set
dopt-0157753c976a14dbb
Main route table
–
Main network ACL
–
Default VPC
No
IPv4 CIDR
192.168.0.0/24
IPv6 pool
–
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Console Mobile App
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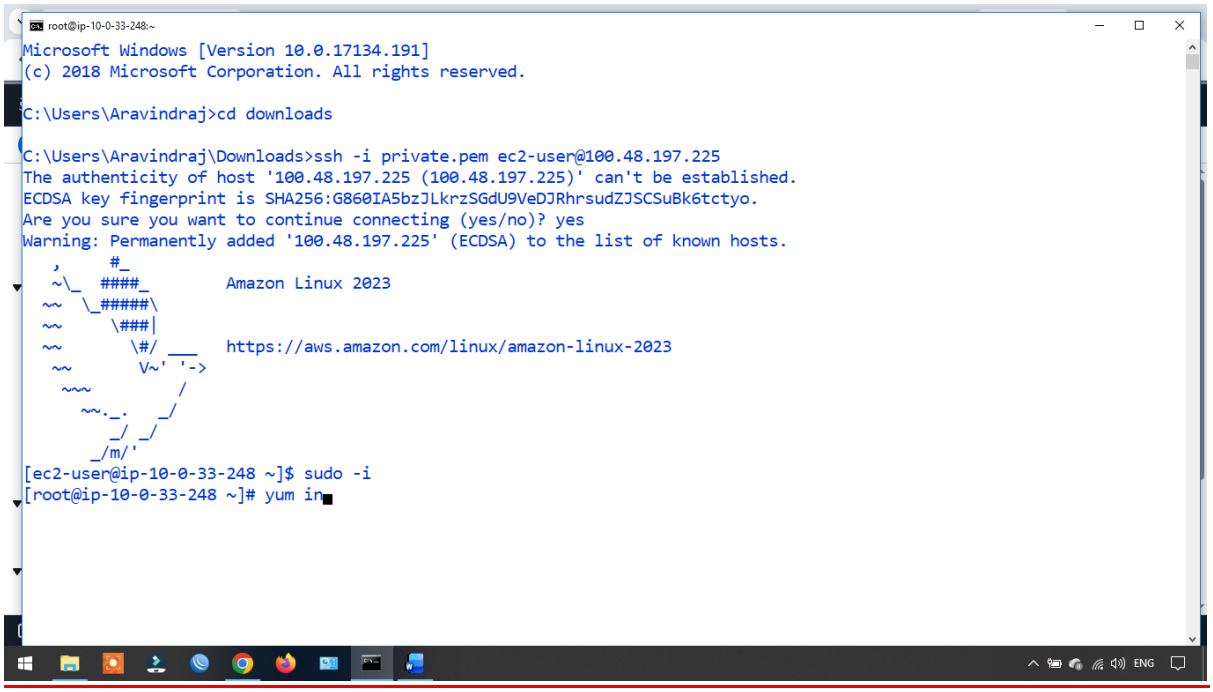

Instances | EC2 | us-east-1
Subnet details | VPC Management
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#subnets:subnetId=subnet-02dc209e60a07a801
Search
[Alt+S]
United States (N. Virginia)
Gokul S (294022385305)
Gokul S
VPC > Subnets
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
You have successfully created 1 subnet: subnet-02dc209e60a07a801
Subnets (1/1) Info
Last updated less than a minute ago
Actions
Create subnet
Find subnets by attribute or tag
Subnet ID : subnet-02dc209e60a07a801
Clear filters
Name
Subnet ID
State
VPC
public 2
subnet-02dc209e60a07a801
Available
vpc-0e53b9fc15a8a3095 | vpc 2
subnet-02dc209e60a07a801 / public 2
Details
Subnet ID
subnet-02dc209e60a07a801
Subnet ARN
arn:aws:ec2:us-east-1:294022385305:subnet/subnet-02dc209e60a07a801
State
Available
Block Public Access
Off
IPv4 CIDR
192.168.0.0/25
Available IPv4 addresses
123
IPv6 CIDR
IPv6 CIDR association ID
VPC
Route table
CloudShell
Feedback
Console Mobile App
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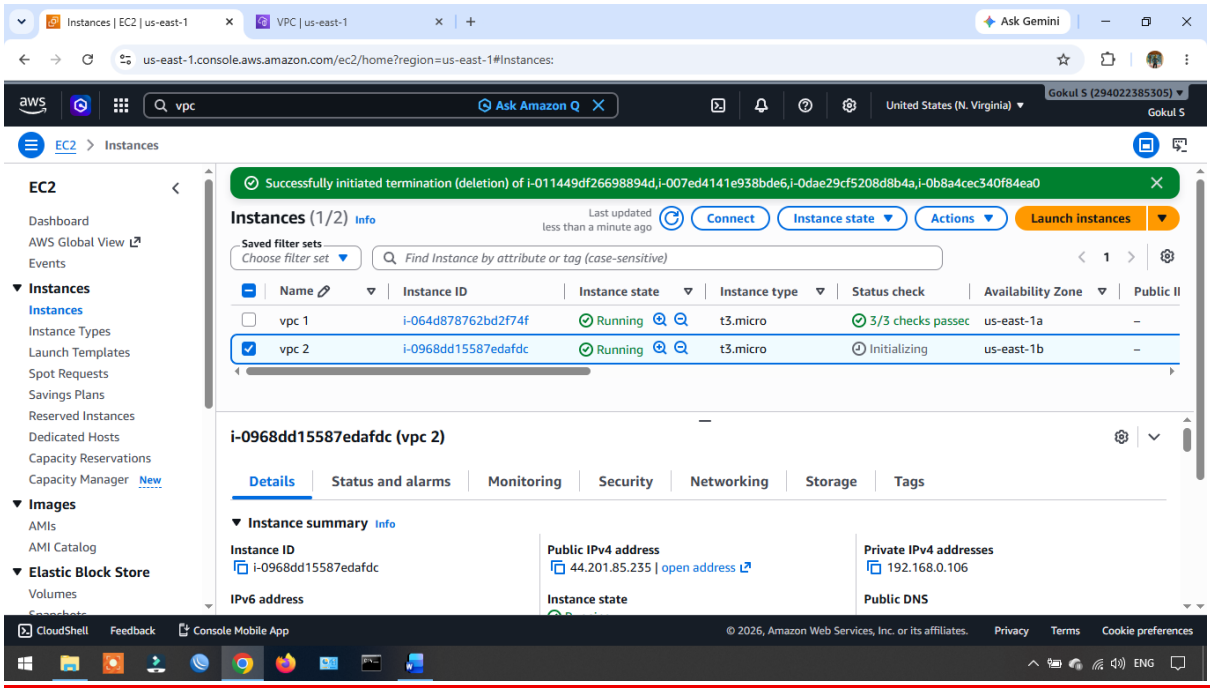

Instances | EC2 | us-east-1
Internet gateway details | VPC
Ask Gemini
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#igws:
Search
[Alt+S]
United States (N. Virginia)
Gokul S (294022385305)
Gokul S
VPC > Internet gateways
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
Internet gateways (1/3) Info
Actions
Create internet gateway
Find internet gateways by attribute or tag
Name
Internet gateway ID
State
VPC ID
Default-VPC-IGW
igw-0d70633d5595dcb14
Attached
vpc-026c40e4766f2ccc4 | Default-VPC
public 1
igw-02efb1ae3e3db7437
Attached
vpc-0337dbbcd6fb4d389 | vpc 1
public 2
igw-03dac5e4ac65b33d0
Attached
vpc-0e53b9fc15a8a3095 | vpc 2
igw-03dac5e4ac65b33d0 / public 2
Details
Tags
Details
Internet gateway ID
igw-03dac5e4ac65b33d0
State
Attached
VPC ID
vpc-0e53b9fc15a8a3095 | vpc 2
Owner
294022385305
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Console Mobile App
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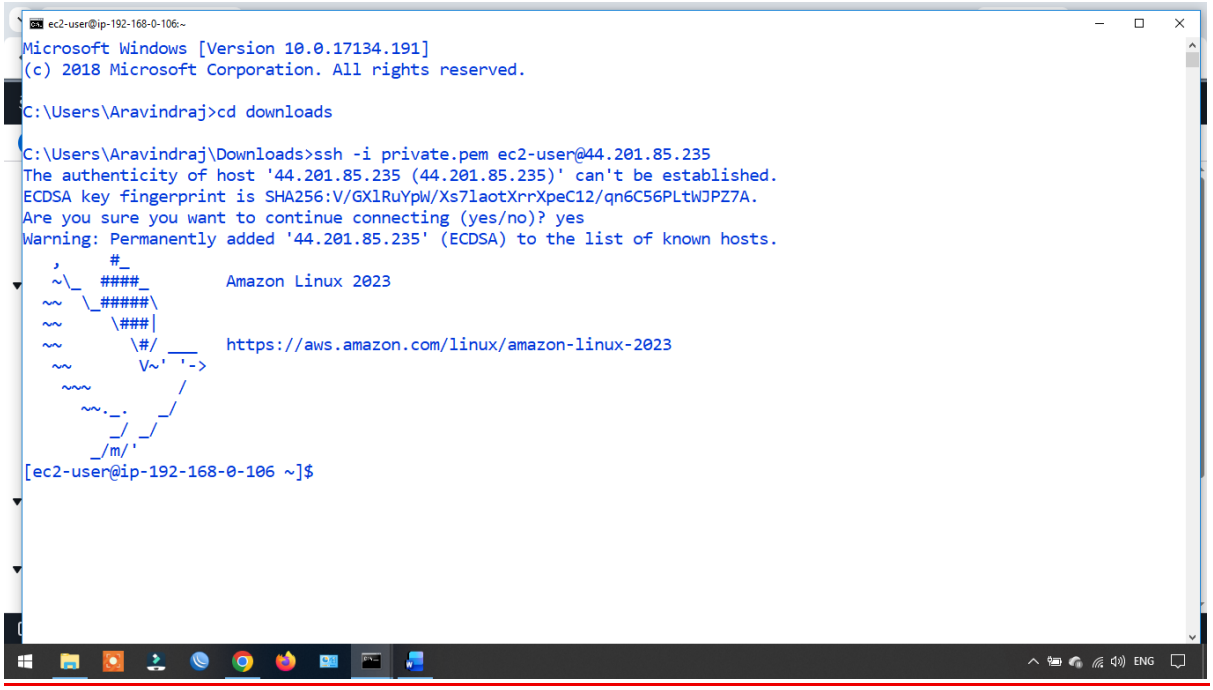


Task2: Create Linux Instance & Install Apache Webserver (EC2) on newly created 2 different VPC's:




root@ip-10-0-33-248:~
Microsoft Windows [Version 10.0.17134.191]
(c) 2018 Microsoft Corporation. All rights reserved.
C:\Users\Aravindraj>cd downloads
C:\Users\Aravindraj\Downloads>ssh -i private.pem ec2-user@100.48.197.225
The authenticity of host '100.48.197.225 (100.48.197.225)' can't be established.
ECDSA key fingerprint is SHA256:G860IA5bzJLkrzSGdU9VeDJRhrsudZJSCSuBk6tctyo.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '100.48.197.225' (ECDSA) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-0-33-248 ~]$ sudo -i
[root@ip-10-0-33-248 ~]# yum in

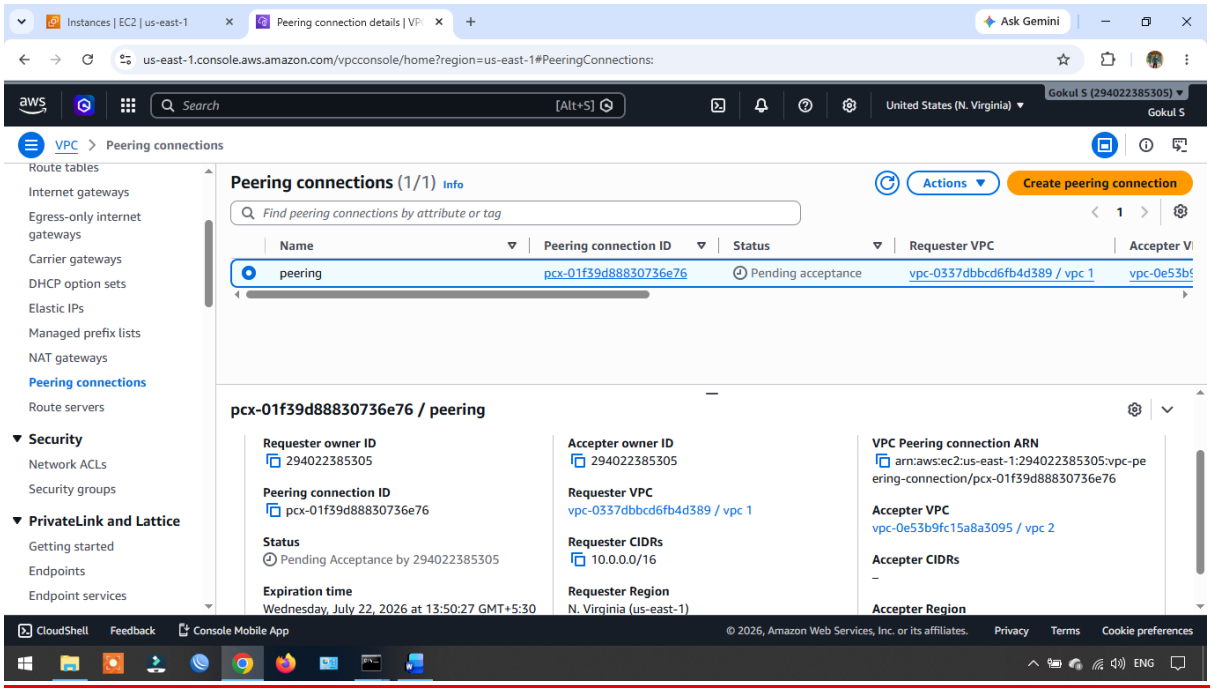

Instances | EC2 | us-east-1
VPC | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
vpc
Ask Amazon Q
United States (N. Virginia)
Gokul S
EC2 > Instances
Successfully initiated termination (deletion) of i-011449df26698894d,i-007ed4141e938bde6,i-0dae29cf5208d8b4a,i-0b8a4cec340f84ea0
Instances (1/2) Info
Connect
Instance state
Actions
Launch instances
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
vpc 1
i-064d878762bd2f74f
Running
t3.micro
3/3 checks passed
us-east-1a
vpc 2
i-0968dd15587edafdc
Running
t3.micro
Initializing
us-east-1b
i-0968dd15587edafdc (vpc 2)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Instance ID
i-0968dd15587edafdc
Public IPv4 address
44.201.85.235 | open address
Private IPv4 addresses
192.168.0.106
IPv6 address
Instance state
Public DNS
CloudShell
Feedback
Console Mobile App
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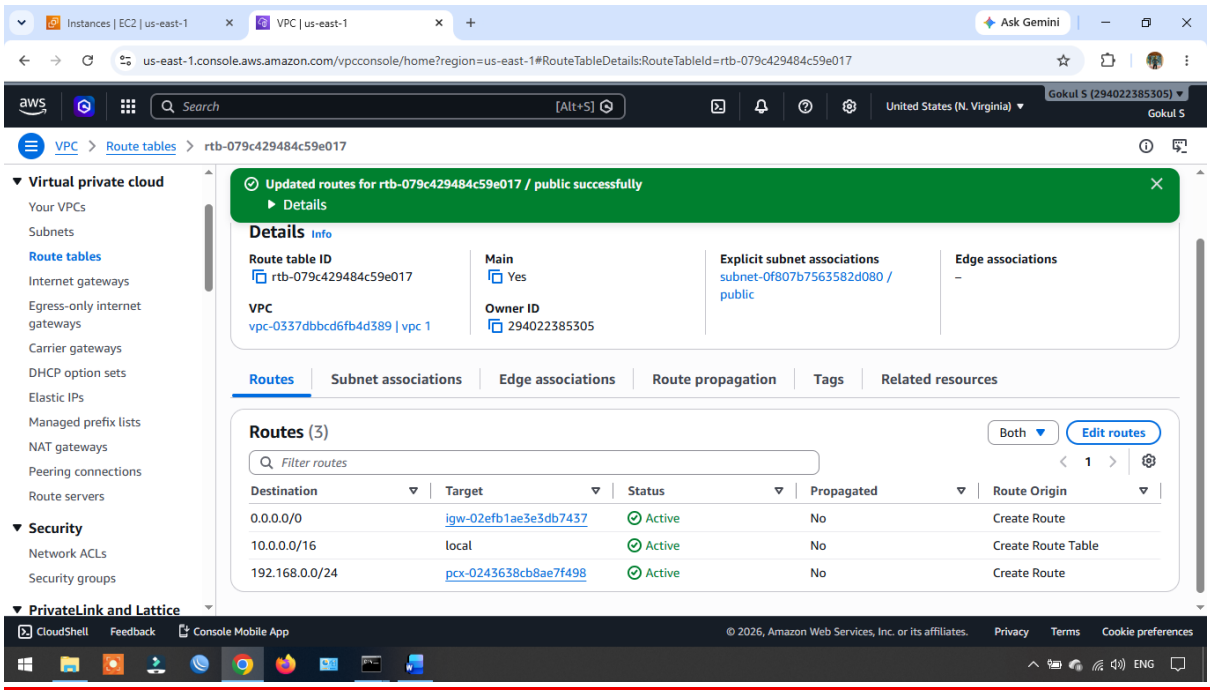


Task3: From VPC1 Linux Instance try to access VPC2 Linux Instance (EC2) by using private IP address ,Is possible to access or not?

Yes ,by peering connection it is possible.

Task4: Configure VPC peering for VPC1 to VPC2 in the Same region with appropriate route table configuration.


Peering connection details | VPC
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#PeeringConnections:
VPC > Peering connections
Peering connections (1/1) Info
Actions
Create peering connection
Find peering connections by attribute or tag
Name
Peering connection ID
Status
Requester VPC
peering
pcx-01f39d88830736e76
Pending acceptance
vpc-0337dbbcd6fb4d389 / vpc 1
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
pcx-01f39d88830736e76 / peering
Requester owner ID
294022385305
Peering connection ID
pcx-01f39d88830736e76
Status
Pending Acceptance by 294022385305
Expiration time
Wednesday, July 22, 2026 at 13:50:27 GMT+5:30
Accepter owner ID
294022385305
Requester VPC
vpc-0337dbbcd6fb4d389 / vpc 1
Requester CIDRs
10.0.0.0/16
Requester Region
N. Virginia (us-east-1)
VPC Peering connection ARN
arn:aws:ec2:us-east-1:294022385305:vpc-peering-connection/pcx-01f39d88830736e76
Accepter VPC
vpc-0e53b9fc15a8a3095 / vpc 2
Accepter CIDRs
Accepter Region
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VPC | us-east-1
us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTableDetails:RouteTableId=rtb-079c429484c59e017
VPC > Route tables > rtb-079c429484c59e017
Updated routes for rtb-079c429484c59e017 / public successfully
Details
Details Info
Route table ID
rtb-079c429484c59e017
Main
Yes
Explicit subnet associations
subnet-0f807b7563582d080 / public
Edge associations
VPC
vpc-0337dbbcd6fb4d389 | vpc 1
Owner ID
294022385305
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (3)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0
igw-02efb1ae3e3db7437
Active
No
Create Route
10.0.0.0/16
local
Active
No
Create Route Table
192.168.0.0/24
pcx-0243638cb8ae7f498
Active
No
Create Route
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
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Task5: Now test the above task

```
C:\Users\starj\Downloads>ssh -i linux-pw.pem ec2-user@3.234.249.126
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Mon May 11 13:46:20 2026 from 10.0.100.155
[ec2-user@ip-10-0-100-155 ~]$ sudo -i
[root@ip-10-0-100-155 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
     Active: active (running) since Mon 2026-05-11 13:25:21 UTC; 30min ago
       Docs: man:httpd.service(8)
   Main PID: 25669 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1014)
     Memory: 13.4M
        CPU: 1.630s
     CGroup: /system.slice/httpd.service
             ├─25669 /usr/sbin/httpd -DFOREGROUND
             ├─25694 /usr/sbin/httpd -DFOREGROUND
             ├─25695 /usr/sbin/httpd -DFOREGROUND
             ├─25696 /usr/sbin/httpd -DFOREGROUND
             └─25704 /usr/sbin/httpd -DFOREGROUND

May 11 13:25:20 ip-10-0-100-155.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
May 11 13:25:21 ip-10-0-100-155.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
```

```
May 11 13:25:20 ip-10-0-100-155.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
May 11 13:25:21 ip-10-0-100-155.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
May 11 13:25:21 ip-10-0-100-155.ec2.internal httpd[25669]: Server configured, listening on: port 80
[root@ip-10-0-100-155 ~]# ls
linux-pw.pem
[root@ip-10-0-100-155 ~]# ssh -i linux-pw.pem ec2-user@192.168.81.60
The authenticity of host '192.168.81.60 (192.168.81.60)' can't be established.
ED25519 key fingerprint is SHA256:2T/g0HM8k/piJ95eFgSz/IYPbjQqgp5dlSw70pgjZ60.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:3: 13.222.232.162
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.81.60' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Mon May 11 13:48:17 2026 from 3.234.249.126
[ec2-user@ip-192-168-81-60 ~]$
```